Data, summaries and distances

Shared notation: N = observations, M = attributes; i, ℓ index observations, j, m attributes, c classes, k components/clusters/folds as stated; x_i = input row, x_{ij} = its entry, y_i = target, $\hat{}$ = estimated/predicted value.

Data matrix and standardization

Purpose: Put attributes on comparable scales before distances, PCA or regularized models.

$$X \in \mathbb{R}^{N\times M}, \qquad \tilde{x}_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j},$$
$$\bar{x}_j = \frac{1}{N}\sum_i x_{ij}, \qquad s_j^2 = \frac{1}{N-1}\sum_i (x_{ij} - \bar{x}_j)^2.$$

Symbols: $X \in \mathbb{R}^{N\times M}$ is the data matrix; $\mathbb{R}$ means real values; $\bar{x}_j$ and s_j are mean and sample standard deviation of attribute j; $\tilde{X}$ is standardized data.
How to read it: $\tilde{x}_{ij} > 0$ means observation i is above the mean of attribute j, measured in standard deviations.

Covariance, correlation and distance

Purpose: Measure joint variation or similarity between observations.

$$\hat{\Sigma} = \frac{1}{N-1}\tilde{X}^\top \tilde{X}, \qquad \rho_{jk} = \frac{\mathrm{Cov}(x_j, x_k)}{s_j s_k},$$
$$d(x_i, x_\ell) = \sqrt{\sum_m (x_{im} - x_{\ell m})^2}.$$

Symbols: $\hat{\Sigma}$ = sample covariance/correlation matrix of standardized attributes; ρ_{jk} = correlation of columns j, k; Cov = covariance; $d(x_i, x_\ell)$ = Euclidean distance between rows. **Means:** covariance/correlation compare columns; distance compares rows. For standardized X, the covariance matrix equals the correlation matrix.
How to read a matrix: diagonal correlation is 1; sign gives direction of linear relation and $|\rho|$ its strength. Distance matrices have zero diagonal and large entries for separated observations.

$$d_1(x, z) = \sum_m |x_m - z_m|, \qquad d_\infty(x, z) = \max_m |x_m - z_m|.$$

Symbols/use: x, z = two vectors; d_1 is cityblock distance; d_∞ is maximum-coordinate distance. Use the named distance in neighbour/density calculations.

Plots and summary objects

Object	How to read it
Histogram/density	Height shows where values concentrate; width/tails show spread and extremes.
Boxplot	Middle line is median; box is middle 50%; whiskers/points show range or unusual values.
Scatter plot	Tilt indicates correlation sign; narrow cloud indicates strong linear relation.
Scatter matrix	Panel (j, k) compares two attributes; match signs to a proposed correlation matrix.

Do this: identify variables and scale; read direction/spread; then match to numerical summaries.

Regression loss

$$e_i = y_i - \hat{y}_i, \quad \mathrm{SSE} = \sum_i e_i^2, \quad \mathrm{MSE} = \frac{1}{N}\sum_i e_i^2,$$
$$\mathrm{RMSE} = \sqrt{\mathrm{MSE}}.$$

Symbols/means: e_i = residual for case i; $y_i, \hat{y}_i$ = true and predicted response; SSE = summed squared error; MSE = its mean; RMSE = its square root in the unit of y. Smaller test/CV RMSE indicates better prediction.

PCA and SVD

Components, scores and reconstruction

Purpose: Replace correlated attributes by orthogonal directions ordered by variation.

$$\tilde{X} = USV^\top, \qquad V = [v_1, \dots, v_M],$$
$$Z = \tilde{X}V = US, \qquad z_{ik} = \tilde{x}_i^\top v_k.$$
$$z_i = \tilde{x}_i V_K, \qquad \hat{\tilde{x}}_i = z_i V_K^\top.$$

Symbols: U = left singular-vector matrix; S = diagonal singular-value matrix; V = loading directions; column v_k is PC k; Z = score matrix; z_i = PC coordinates of row i; V_K retains K PCs; $\hat{\tilde{x}}_i$ = reconstructed standardized row; s_k = singular value.
How to read it: large $|v_{mk}|$ means attribute m defines PC k strongly. A positive score lies in the direction of positive loadings. Reversing all signs on one PC changes neither the PCA representation nor distances.

Explained variance

$$\lambda_k = \frac{s_k^2}{N-1}, \quad \mathrm{EVR}_k = \frac{s_k^2}{\sum_j s_j^2}, \quad \mathrm{cumEVR}(K) = \frac{\sum_{k=1}^K s_k^2}{\sum_j s_j^2}.$$

Symbols/means: λ_k = variance along PC k; EVR_k = its fraction of total variance; $\mathrm{cumEVR}(K)$ = fraction retained by the first K PCs.
Do this: square singular values; divide by their total; accumulate in PC order; choose the smallest K reaching the target.

Read PCA output and plots

Object	Interpretation
Loading column v_k	Read down the column: sign and magnitude connect original attributes to PC k.
Score plot	Each point is one observation; nearby points have similar retained attribute profiles.
Loading plot	Arrows/points are attributes; similar direction means similar contribution pattern.
Biplot	Observations projected along a variable direction tend to have high values of that variable.

Do this for a projection: take the requested column of V; multiply corresponding standardized attribute values; add terms to obtain z_{ik}.

Norm and cosine

$$\|x\|_2 = \sqrt{x^\top x}, \qquad \|X\|_F^2 = \sum_{ij} x_{ij}^2 = \sum_k s_k^2,$$
$$\cos(x, y) = \frac{x^\top y}{\|x\|\|y\|}.$$

Symbols/means: $\|\cdot\|_2$ = vector length; $\|X\|_F$ = Frobenius length of all matrix entries; $x^\top y$ = dot product; $\cos(x, y)$ reads similar direction, while Euclidean distance also responds to magnitude.

Probability and Bayes classification

Conditional probability and Bayes

Purpose: Update class probabilities after seeing attributes.

$$P(A|B) = \frac{P(A, B)}{P(B)}, \quad P(A) = \sum_c P(A|c)P(c),$$
$$P(y = c|x) = \frac{P(x|y = c)P(y = c)}{\sum_{c'} P(x|y = c')P(y = c')}.$$

Symbols: $P(A|B)$ = probability of event A given B; x = observed attribute vector; y = label; $P(y = c)$ = class prior; $P(x|y = c)$ = likelihood; $P(y = c|x)$ = posterior; c' indexes candidate classes.
Do this: compute likelihood times prior for every class; sum scores; divide each score by the sum; select largest posterior.

Counts and Naive Bayes

Purpose: Classify from categorical or binary attribute tables.

$$\hat{P}(y = c|x = a) = \frac{\#\{i : y_i = c, x_i = a\}}{\#\{i : x_i = a\}},$$
$$\mathrm{score}(c) = P(y = c)\prod_{m=1}^M P(x_m|y = c).$$
$$P(y = c|x) = \frac{\mathrm{score}(c)}{\sum_{c'} \mathrm{score}(c')}.$$

Symbols/means: $\#\{\cdot\}$ = number of matching rows; a = specified attribute value; x_m = attribute m; score(c) = unnormalized posterior. Naive Bayes factorizes the likelihood by conditional independence within class.
How to read a probability table: use the class row/column matching c for each factor; posterior values are normalized class memberships and add to 1.

Gaussian density classifier

Purpose: Assign continuous observations using class-specific centres and spreads.

$$\mathcal{N}(x|\mu,\Sigma) = \frac{\exp[-\frac{1}{2}(x-\mu)^\top\Sigma^{-1}(x-\mu)]}{(2\pi)^{M/2}|\Sigma|^{1/2}},$$

$$\hat{y} = \arg\max_c\{\mathcal{N}(x|\mu_c,\Sigma_c)P(c)\}.$$

$$\mathcal{N}(x|\mu,\sigma^2) = \frac{1}{\sqrt{2\pi}\sigma}\exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right].$$

Symbols: $\mathcal{N}$ = Gaussian density; μ,Σ = centre and covariance, with subscript c for a class; $|\Sigma|$ = determinant; Σ^{-1} = inverse covariance; σ = univariate standard deviation; $\hat{y}$ = predicted class; $\arg\max_c$ selects the largest class score.
How to read density curves: points near μ receive high density; larger variance creates a wider, lower curve. Density is a height, whereas posterior is a class probability.

Log scores

$$\log\operatorname{score}(c) = \log P(c) + \sum_m \log P(x_m|c).$$

Symbols/use: log = logarithm; m indexes attributes. Use when many small factors are multiplied; the largest log-score gives the same class decision.

Prediction models and boundaries

Linear and ridge regression

Purpose: Predict a continuous response as a weighted sum of attributes.

$$\bar{x} = [1, x_1, \ldots, x_M]^\top, \quad X_a = [\mathbf{1}, X], \quad \hat{y} = \bar{x}^\top w,$$

$$w_{\text{LS}} = (X_a^\top X_a)^{-1}X_a^\top y, \qquad D = \operatorname{diag}(0,1,\ldots,1),$$

$$w_\lambda = (X_a^\top X_a + \lambda D)^{-1}X_a^\top y.$$

Symbols: $\bar{x}$ = input with leading constant 1; $X_a = [\mathbf{1}, X]$ = design matrix with intercept column; w = coefficient vector; w_{LS} = least-squares fit; w_λ = ridge fit; w_0 = intercept; w_j = change per unit x_j; y = response vector; D excludes the intercept from penalty; λ = shrinkage strength.
How to read weights: for standardized attributes, larger $|w_j|$ indicates stronger linear influence on predictions. Increasing λ pulls non-intercept weights toward zero.

$$E_\lambda(w) = \sum_i (y_i - \bar{x}_i^\top w)^2 + \lambda\sum_{j=1}^M w_j^2.$$

Symbols/do this: E_λ = penalized training cost. Compute residual squares; add the penalty from non-intercept coefficients; use validation error to choose λ.

Logistic regression

Purpose: Map a weighted input score to probability of the positive class.

$$a = \bar{x}^\top w, \qquad \hat{p} = P(y=1|x) = \sigma(a) = \frac{1}{1+e^{-a}},$$

$$\hat{y} = \mathbb{I}[\hat{p} \geq \tau],$$

$$\log\frac{\hat{p}}{1-\hat{p}} = \bar{x}^\top w.$$

Symbols: a = linear score/log-odds; $\hat{p}$ = predicted positive-class probability; σ = sigmoid; τ = decision threshold; $\mathbb{I}[\cdot] = 1$ when its condition holds and 0 otherwise. **How to read it:** positive w_j raises the log-odds as x_j increases. At $\tau = 0.5$, the decision boundary satisfies $\bar{x}^\top w = 0$.
Do this: insert attributes including intercept; compute a; convert with sigmoid; compare with stated threshold and class definition.

K-nearest neighbours

Purpose: Predict from nearby training observations in input space.

$$\hat{y}_{\text{reg}}(x) = \frac{1}{K}\sum_{i\in N_K(x)} y_i,$$

$$\hat{y}_{\text{class}}(x) = \arg\max_c \sum_{i\in N_K(x)} \mathbb{I}[y_i = c].$$

Symbols: $N_K(x)$ = the K closest training points to new input x; $\hat{y}_{\text{reg}}$ = neighbour-response mean; $\hat{y}_{\text{class}}$ = class with most neighbour votes; $\mathbb{I}[y_i = c]$ counts votes for class c.
How to read it: small K produces local, irregular boundaries; larger K averages more neighbours and yields smoother predictions.

Multiclass scores and boundary plots

$$p_c = \frac{e^{a_c}}{\sum_j e^{a_j}}, \qquad \hat{y} = \arg\max_c p_c.$$

Symbols: a_c = score for class c; p_c = softmax class probability; j in the denominator runs over classes; $\hat{y}$ = class with greatest probability.

Boundary appearance	Model reading
Straight line/plane	Linear score, e.g. logistic regression.
Axis-aligned regions	A decision tree splits one attribute per node.
Irregular local cells	KNN prediction follows nearby observations.
Smooth curved region	Nonlinear features or neural network.

Means: softmax converts one score per class to probabilities summing to 1.

Validation and classifier output

Hold-out and cross-validation

Purpose: Estimate predictive error on observations not used for fitting.

$$E_T = \frac{1}{|T|}\sum_{i\in T} L(y_i, \hat{f}(x_i)), \qquad E_{\text{CV}} = \frac{1}{K}\sum_{k=1}^K E_k.$$

Symbols: T = held-out/test set; $|T|$ = its number of rows; $L(y_i, \hat{f}(x_i))$ = loss of fitted predictor $\hat{f}$ on row i; E_T = test error; E_k = error on validation fold k; K = number of folds; E_{CV} = fold-average error.
How to read a CV table: select hyperparameters by inner/validation error; outer test-fold error estimates the selected procedure's generalization error.

Stage	Data use
Fit/preprocess	Learn scaling, features and weights only from the training portion.
Select model	Compare candidates on validation/inner folds.
Report error	Evaluate selected procedure on held-out/outer folds.

Confusion matrix and metrics

Purpose: Summarize binary decisions relative to the chosen positive class.

	$\hat{y} = +$	$\hat{y} = -$
$y = +$	TP	FN
$y = -$	FP	TN

$$\text{Acc} = \frac{TP+TN}{N}, \qquad \text{Recall} = \text{TPR} = \frac{TP}{TP+FN},$$

$$\text{FPR} = \frac{FP}{FP+TN}, \qquad \text{Precision} = \frac{TP}{TP+FP},$$

$$\text{Specificity} = \frac{TN}{TN+FP}, \quad F_1 = \frac{2\ \text{Precision Recall}}{\text{Precision} + \text{Recall}}.$$

Symbols: TP, TN = correct positive/negative decisions; FP, FN = false positive/negative decisions; N = all classified cases; TPR = true-positive rate; FPR = false-positive rate; F_1 = harmonic mean of precision and recall. **How to read it:** recall asks "which actual positives were found?"; precision asks "which predicted positives were correct?"; FPR measures negative cases incorrectly flagged positive.
Do this: define the positive class; place counts in the table; use the denominator matching the requested interpretation.

ROC curves and thresholds

$$\text{ROC}(\tau) = (\text{FPR}(\tau), \text{TPR}(\tau)), \qquad \text{AUC} = \text{area under ROC}.$$

Symbols: τ = positive-score threshold; ROC = curve of FPR against TPR as τ varies; AUC = its area. **How to read it:** lowering τ predicts more positives, moving from $(0,0)$ toward $(1,1)$. A curve near the upper-left separates classes well; random rankings give AUC about 0.5.
Do this: sort predicted positive scores from high to low; add one case at a time as positive; update TPR/FPR; plot or identify the resulting point.

Paired comparison: McNemar

Purpose: Compare two classifiers on the same evaluated observations.

	B correct	B wrong
A correct	n_{11}	n_{12}
A wrong	n_{21}	n_{22}

$$\hat{\theta} = \frac{n_{12} - n_{21}}{N}.$$

$$p = 2F_{\mathrm{Binom}(n_{12}+n_{21},\,1/2)}(\min(n_{12}, n_{21}))\,.$$

Symbols: A, B = the two classifiers; n_{rs} = counts in the shown paired table; $\hat{\theta}$ = estimated accuracy difference; $F_{\mathrm{Binom}(n,1/2)}$ = binomial CDF for n disagreements under equal performance; p = two-sided p-value. **How to read it:** only disagreements n_{12}, n_{21} determine whether accuracies differ; positive $\hat{\theta}$ means A has higher accuracy under the table convention above. A small p indicates evidence of different performance.

Paired regression loss difference

Purpose: Decide whether two regressors differ on paired test losses.

$$z_i = L_A(i) - L_B(i), \quad \bar{z} = \frac{1}{N}\sum_i z_i, \quad s_z^2 = \frac{1}{N-1}\sum_i (z_i - \bar{z})^2,$$

$$\mathrm{CI}_{1-\alpha}: \quad \bar{z} \;\pm\; t_{1-\alpha/2, N-1}\frac{s_z}{\sqrt{N}}.$$

Symbols: $L_A(i), L_B(i)$ = model losses on test case i; z_i = paired loss difference; $\bar{z}, s_z$ = its sample mean and standard deviation; α = error level; $t_{1-\alpha/2,N-1}$ = Student-t critical value; CI = confidence interval. **How to read it:** $\bar{z} < 0$ favours model A when lower loss is better; a confidence interval containing 0 does not demonstrate a difference at that confidence level.

Trees, ensembles and neural networks

Decision tree split

Purpose: Divide input space into regions with homogeneous class labels or responses.

$$I_{\mathrm{error}} = 1 - \max_c p_c, \quad I_{\mathrm{Gini}} = 1 - \sum_c p_c^2,$$

$$I_{\mathrm{ent}} = -\sum_c p_c \log_2 p_c,$$

$$\mathrm{gain} = I(\mathrm{parent}) - \sum_v \frac{N_v}{N} I(\mathrm{child}\ v).$$

Symbols: $I_{\mathrm{error}}, I_{\mathrm{Gini}}, I_{\mathrm{ent}}$ = node impurity measures; p_c = class fraction in a node; v indexes children; N_v/N = child size fraction; gain = impurity removed by a split.
How to read a tree: follow threshold branches from root to leaf; a classification leaf returns its majority/class distribution, a regression leaf its response average. Splits give axis-aligned regions in a boundary plot.
Do this: count labels in each candidate child; compute weighted child impurity; use the split with largest gain.

AdaBoost

Purpose: Combine weak classifiers while emphasizing observations missed in previous rounds.

$$\epsilon_t = \sum_i q_i^{(t)} \mathbb{I}[h_t(x_i) \neq y_i], \qquad \alpha_t = \frac{1}{2}\log\frac{1-\epsilon_t}{\epsilon_t},$$

$$q_i^{(t+1)} \propto q_i^{(t)} \exp\big(\alpha_t \mathbb{I}[h_t(x_i) \neq y_i]\big),$$

$$H(x) = \mathrm{sign} \sum_t \alpha_t h_t(x).$$

Symbols/means: t = boosting round; $q_i^{(t)}$ = weight of row i; $h_t(x)$ = weak classifier returning a signed class; ϵ_t = weighted error; α_t = vote weight; $H(x)$ = final signed class. Missed observations receive more relative weight next.
Do this: compute ϵ_t from current weights; find α_t; update and normalize observation weights; combine signed votes.

Neural-network forward pass

Purpose: Form nonlinear predictions by transforming weighted sums through hidden units.

$$a_j = w_{0j}^{(1)} + \sum_m w_{mj}^{(1)} x_m, \quad h_j = g(a_j), \quad \hat{y} = w_0^{(2)} + \sum_j w_j^{(2)} h_j.$$

$$\mathrm{ReLU}(a) = \max(0, a), \qquad \tanh(a) \in [-1, 1],$$

$$\sigma(a) = \frac{1}{1+e^{-a}} \in (0, 1).$$

Symbols: a_j = pre-activation of hidden unit j; g = activation; h_j = hidden output; $w_{mj}^{(1)}$ = input-to-hidden weight; $w_j^{(2)}$ = hidden-to-output weight; $w_{0j}^{(1)}, w_0^{(2)}$ = biases; $\hat{y}$ = network output; ReLU, $\tanh, \sigma$ = shown activations.
How to read a diagram/table: gather incoming weights for each hidden unit first; apply its activation; then use outgoing weights to calculate prediction or final class probability.
Do this: calculate all a_j; apply g; calculate output; apply final sigmoid/threshold only when specified for classification.

Clustering, density and outliers

K-means and centroids

Purpose: Partition observations into K centre-based clusters.

$$\min_{\{C_k, \mu_k\}} \sum_{k=1}^{K} \sum_{i \in C_k} \|x_i - \mu_k\|^2, \qquad \mu_k = \frac{1}{|C_k|}\sum_{i \in C_k} x_i.$$

Symbols: K = number of clusters; C_k = set of rows assigned to cluster k; $|C_k|$ = its size; μ_k = centroid/mean vector; $\|\cdot\|^2$ = squared Euclidean distance minimized by K-means.
How to read output: each observation belongs to its nearest centroid; a centroid table is a mean attribute profile for each cluster.
Do this: compute distances to centroids; assign nearest; recompute cluster means; repeat assignments if requested.

Hierarchical clustering and dendrograms

Linkage	Cluster-to-cluster distance
Single	minimum pair distance; joins through close links.
Complete	maximum pair distance; favours compact groups.
Average	average of all cross-cluster pair distances.
Centroid	distance between cluster means.

How to read a dendrogram: leaves are observations; branch height is merge distance; a horizontal cut gives the connected groups below that height.

Compare clusterings

$$\mathrm{Rand} = \frac{f_{11} + f_{00}}{\binom{N}{2}}, \qquad J = \frac{f_{11}}{f_{11} + f_{10} + f_{01}}.$$

Symbols: $\binom{N}{2}$ = number of observation pairs; f_{11} = together in both clusterings; f_{00} = apart in both; f_{10}, f_{01} = pair-status disagreements; J = Jaccard agreement.
Means: Rand counts both kinds of agreement; Jaccard measures overlap of same-cluster pairs only.

Gaussian mixture model

Purpose: Describe data by weighted Gaussian components with soft memberships.

$$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x|\mu_k, \Sigma_k), \qquad \sum_k \pi_k = 1,$$

$$\gamma_{ik} = \frac{\pi_k \mathcal{N}(x_i|\mu_k, \Sigma_k)}{\sum_j \pi_j \mathcal{N}(x_i|\mu_j, \Sigma_j)}.$$

Symbols: $p(x)$ = mixture density; π_k = weight of component k; $\mathcal{N}(x|\mu_k, \Sigma_k)$ = its Gaussian density; γ_{ik} = responsibility for row i; μ_k, Σ_k = component mean and covariance; j in the denominator indexes components.
How to read output: π_k is component frequency; γ_{ik} is observation i's responsibility for component k and sums to 1 across components. Largest responsibility gives a hard label.

KDE and local density

Purpose: Find regions of low probability density and potential outliers.

$$\hat{p}(x) = \frac{1}{N}\sum_i \mathcal{N}(x|x_i, \sigma^2 I),$$

$$\ell_{\text{LOO}}(\sigma) = \sum_i \log\left[\frac{1}{N-1}\sum_{j\neq i}\mathcal{N}(x_i|x_j, \sigma^2 I)\right].$$

$$\text{dens}(x_i, K) = \left(\frac{1}{K}\sum_{x_j\in N_i} d(x_i, x_j)\right)^{-1},$$

$$\text{ARD}(x_i, K) = \frac{1}{K}\sum_{x_j\in N_i}\frac{\text{dens}(x_i, K)}{\text{dens}(x_j, K)}.$$

Symbols: $\hat{p}$ = kernel density estimate; σ = Gaussian bandwidth; I = identity matrix; ℓ_{LOO} = leave-one-out log likelihood; $N_i = K$ nearest neighbours of x_i; dens = inverse mean neighbour distance; ARD = average relative density ratio. **How to read it:** small KDE/local density marks isolation; small ARD marks a point less dense than its neighbours. Larger σ smooths a KDE curve.
Do this: exclude the evaluated point for leave-one-out/local-neighbour calculations; compute densities; identify the smallest density or ARD as most outlying.

Associations and sparse similarity

Association rules and Apriori

Purpose: Quantify which items co-occur and form frequent rules.

$$\text{supp}(X) = \frac{\#(X)}{N},$$

$$\text{conf}(X\to Y) = \frac{\text{supp}(X\cup Y)}{\text{supp}(X)},$$

$$\text{lift}(X\to Y) = \frac{\text{conf}(X\to Y)}{\text{supp}(Y)}.$$

Symbols: X = antecedent itemset; Y = consequent itemset; $\#(X)$ = baskets containing X; N = number of baskets; supp = support; conf = confidence; lift compares observed confidence with baseline support of Y. For a rule, X, Y are disjoint.
How to read a rule table: support is prevalence of all rule items; confidence is conditional frequency of Y among baskets with X; lift > 1 indicates more co-occurrence than expected from Y's frequency.
Do this: count X, Y and $X\cup Y$; calculate support/confidence/lift; for Apriori, keep itemsets meeting support and extend only retained sets.

Binary attributes and Jaccard

	$b=1$	$b=0$
$a=1$	f_{11}	f_{10}
$a=0$	f_{01}	f_{00}

$$J(a,b) = \frac{f_{11}}{f_{11}+f_{10}+f_{01}},$$

$$\text{SMC} = \frac{f_{11}+f_{00}}{M}.$$

Symbols/means: a, b = two binary vectors; f_{rs} = number of attributes having values r in a and s in b; M = attribute count; J = Jaccard; SMC = simple matching coefficient. Jaccard ignores shared absences; SMC counts both agreements.
Use when: Jaccard suits sparse baskets or word-presence vectors where common zeros do not show similarity.

Text vectors and cosine

Purpose: Compare documents by weighted word content rather than raw document length.

$$\text{tfidf}_{ij} = \text{tf}_{ij}\log\frac{N}{\text{df}_j}, \qquad \cos(x,y) = \frac{x^\top y}{\|x\|_2\|y\|_2}.$$

Symbols: tf_{ij} = term count/weight for word j in document i; df_j = documents containing word j; N = documents; x, y = tf–idf document vectors; $x^\top y$ = dot product; $\|\cdot\|_2$ = vector length.
How to read it: rare terms receive larger inverse-document-frequency weight; cosine near 1 means documents point in similar word-weight directions.

Choose from the given object

Given in task	**Calculate/read**
Label and class score	Posterior/logistic probability; confusion/ROC for performance.
Numeric response and weights	Regression prediction, residual/RMSE and validation error.
No label; centroids/tree of merges	K-means assignment or dendrogram cut.
Density or neighbour table	GMM responsibility, KDE/local density or ARD outlier score.
Item/word table	Association metrics, Jaccard or cosine similarity.

Numerical output meanings

Quantity	**Reading rule**
Probability/responsibility	Between 0 and 1; competing class/component values sum to 1.
EVR/AUC/metric	EVR sums to 1 across PCs; AUC closer to 1 means stronger ranking.
Density	Nonnegative local height; compare values to locate low-density observations.
Similarity	Cosine/Jaccard large means more similar under their respective representation.